

Meridian Parts

European service-hub expansion

Board recommendation: fund CZE, ESP through a gated 90-day validation, reserving €225,000 of year-zero capex and 5 FTE. The selected pair produces €198,099 of base-case annual incremental contribution, with a 1.14-year simple payback.

Decision date: 27 September 2026

Scope: 2025 shipped sales and returns received by 31 January 2026; six European markets. Inputs are synthetic client exports and assumptions, with archived official World Bank and ECB snapshots identified in the source register.

Prepared for the Meridian Parts board

1. Executive decision

Approve a **gated two-hub plan in Czechia (CZE) and Spain (ESP)**, but release capex only after the 90-day validation gates. The pair is the highest-ranked feasible alternative on the stated base-case planning metric: €198,099 annual incremental contribution after recurring fixed cost, €225,000 capex, 5 FTE and 1.14 years simple payback.

The recommendation is robust to the defined joint stress in sign but not in magnitude: stress contribution is €32,180 and payback extends to 6.99 years. Low/base/high volume cases produce €64,619, €198,099 and €331,580. The pair's calculated volume-uplift break-even is 2.74%; the board should treat that as a gate, not as evidence that demand will materialize.

Selected pair	Capex	FTE	Low	Base	High	Stress	Base payback
CZE+ESP	€225,000	5	€64,619	€198,099	€331,580	€32,180	1.14 years

Trade-off. The strongest base-case alternative is POL+ESP: €186,329 base contribution, €240,000 capex, 6 FTE, 1.29 years payback and €29,264 stress contribution. NLD+ESP is the more conservative stress-oriented alternative (€107,630 stress contribution) but gives up €31,168 of base-case annual contribution. This is a preference trade-off, not a claim that one scenario is true.

Board ask: authorize discovery, lane-cost validation, and hiring/partner selection work inside the 90-day plan; make the capex release contingent on measured service economics and the volume gate.

2. Evidence, scope and accounting

The analysis uses the frozen source room collected from the TOOLING.md URL. Synthetic client files define the transaction grain, revisions, costs, hub options and scenario policy [S01–S08]. Public context is taken from archived World Bank responses for population and GDP per capita [S09–S10] and an archived ECB historical reference-rate file [S11–S12]. The source register preserves the original URL, retrieval vintage, local hash, units, period and public/synthetic status.

The 2025 base is defined by **shipped_at** in calendar 2025. Returns are included when the selected return revision is linked to an eligible order and received by 31 January 2026 inclusive; refund and returned-unit amounts are allocated to the original sale month. For PLN and CZK, gross sale and refund conversion uses the sale month's arithmetic mean of available ECB business-day quotes in local currency units per EUR; EUR uses 1.0. Monetary components are rounded half-up per order to cents before aggregation [S01].

Booked revenue is the order-ledger sale measure: gross sales before refunds, or net sales after refund credits. Cash is not directly observed in the exports; settlement timing, payment fees, taxes, chargebacks and cash collection are therefore not estimated. Contribution is the analytical operating measure: net sales less net COGS and fulfillment. It excludes hub capex and recurring hub fixed cost, which are modeled separately in scenario economics.

Evidence state: transaction results are computed from saved synthetic inputs; World Bank and ECB values are archived official observations; hub demand uplift, savings per unit, recurring cost and capex are synthetic assumptions. No interviews or live customer account access were used.

3. 2025 operating diagnosis

The baseline is concentrated in the markets that also rank well in the hub arithmetic. Spain has the highest contribution and margin; Czechia is second on contribution and has the lowest capex/FTE intensity among the top-volume options. This supports prioritizing them for validation, but the observed 2025 ledger is not causal evidence that a local hub created demand.

2025 contribution by market

Country totals; EUR, after refunds, net COGS and fulfillment



Source: computed from canonical shipped orders; source register S03–S08.

Countr y	Orders	Units	Gross	Refunds	Net sales	Net COGS	Fulfillment	Contribution	Margin	Returned units
DEU	209	12,153	€631,245	€5,453	€625,792	€349,980	€13,446	€262,366	41.9%	73
FRA	209	11,880	€640,930	€8,699	€632,231	€341,453	€14,507	€276,271	43.7%	144
NLD	209	12,576	€709,109	€11,376	€697,733	€365,496	€16,593	€315,644	45.2%	144
POL	209	13,163	€776,508	€14,618	€761,890	€386,698	€18,698	€356,494	46.8%	180
CZE	209	13,733	€834,572	€19,265	€815,307	€397,062	€20,910	€397,335	48.7%	288
ESP	209	13,931	€872,259	€21,420	€850,839	€406,230	€22,708	€421,900	49.6%	252

The 2025 country totals reconcile to 72 monthly records (12 months × 6 countries); all six reconciliation checks pass. Contribution margin ranges from 41.9% in Germany to 49.6% in Spain on this accounting definition. Returns are physical units, not return-order counts; restocked units recover only the original unit cost [S01].

4. Data quality and reconciliation

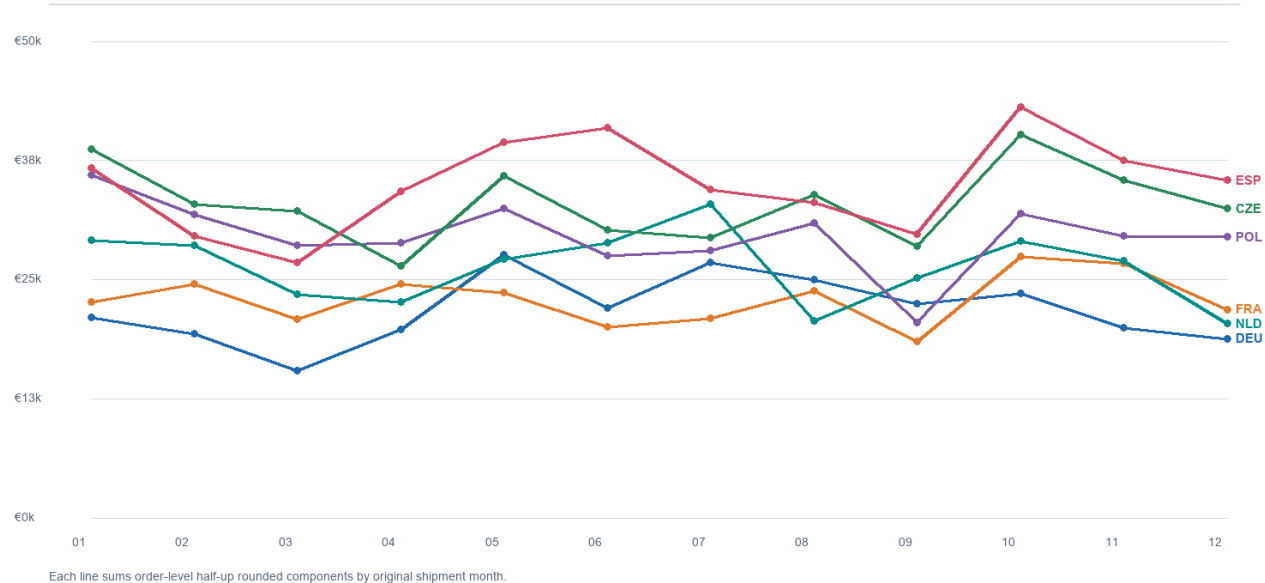
The source room contains deliberate duplicates, corrections and edge cases. The controls below are part of the analysis rather than post-hoc cleanup. Counts are computed from the saved files and can be reproduced from `analysis_outputs/orders_audit.csv` and `returns_audit.csv`.

Issue	Count	Handling
Raw order rows	1,310	Two extract pages combined.
Identical order rows	13	Removed once; not counted as orders.
Order corrections	18	Whole-row replacement; revision 2 selected.
Canonical orders	1,297	Highest revision per order_id.
Cancelled rows	24	Excluded; status not shipped.
Test rows	18	Excluded; is_test=true.
January 2026 shipped rows	1	Excluded from 2025 base.
Zero-price valid shipments	12	Retained; quantity and costs remain in the base.
Raw return rows	264	Return extract combined.
Identical return rows	20	Removed once.
Lower-revision return rows	1	Highest return revision selected.
Included returns	241	Linked and received by cutoff.
Orphan / post-cutoff returns	1 / 1	Orphan quarantined; late return excluded.

No missing 2025 PLN/CZK rate means or missing effective unit costs were detected for eligible orders. Monthly totals use distinct canonical order IDs and are the sum of order-level rounded components. The model does not silently join the orphan return or treat a raw row as an order.

Monthly contribution pattern

2025 contribution by country; EUR



5. Market context and FX

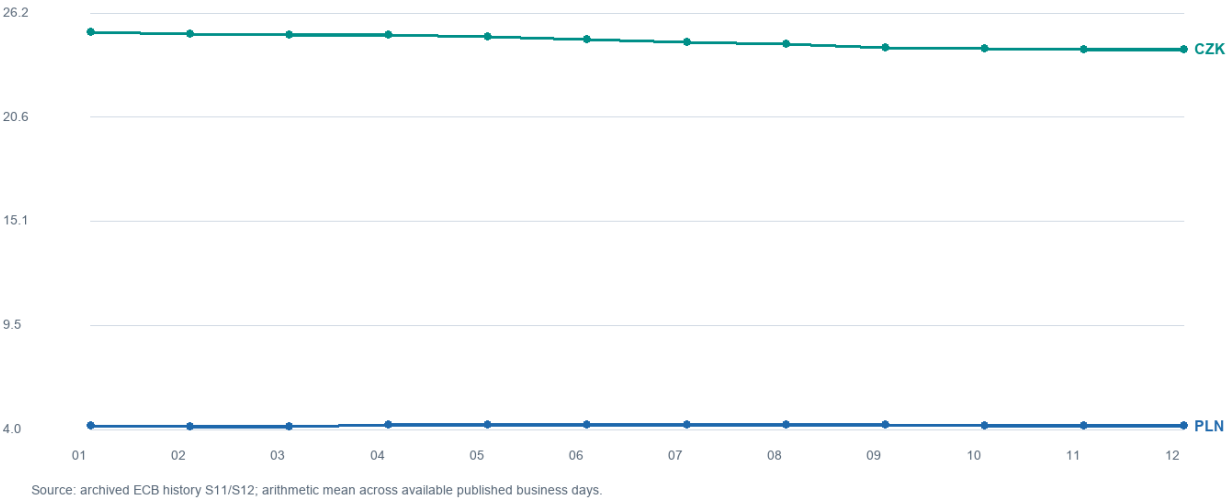
World Bank context shows population change and nominal GDP per capita trends across the six markets. It provides economic context for execution and service design, not direct proof of product demand [S09–S10].

Country	Pop. 2022	Pop. 2024	Change	GDP pc 2022	GDP pc 2024	Change
DEU	83,177,813	83,516,593	+0.4%	\$50,507	\$56,104	+11.1%
FRA	68,184,457	68,551,653	+0.5%	\$40,989	\$46,103	+12.5%
NLD	17,700,982	17,993,485	+1.7%	\$59,123	\$67,465	+14.1%
POL	36,821,749	36,559,233	-0.7%	\$18,891	\$25,104	+32.9%
CZE	10,672,118	10,905,028	+2.2%	\$28,282	\$31,828	+12.5%
ESP	47,786,102	48,848,840	+2.2%	\$30,319	\$35,327	+16.5%

Population increased in Germany, France, Netherlands, Czechia and Spain over 2022–2024 in the archive; Poland declined. Nominal current-US-dollar GDP per capita increased in all six in the archived values, but exchange rates and price effects mean it is not a real-income or demand measure.

2025 ECB monthly reference-rate means

Local currency units per EUR; not transaction FX



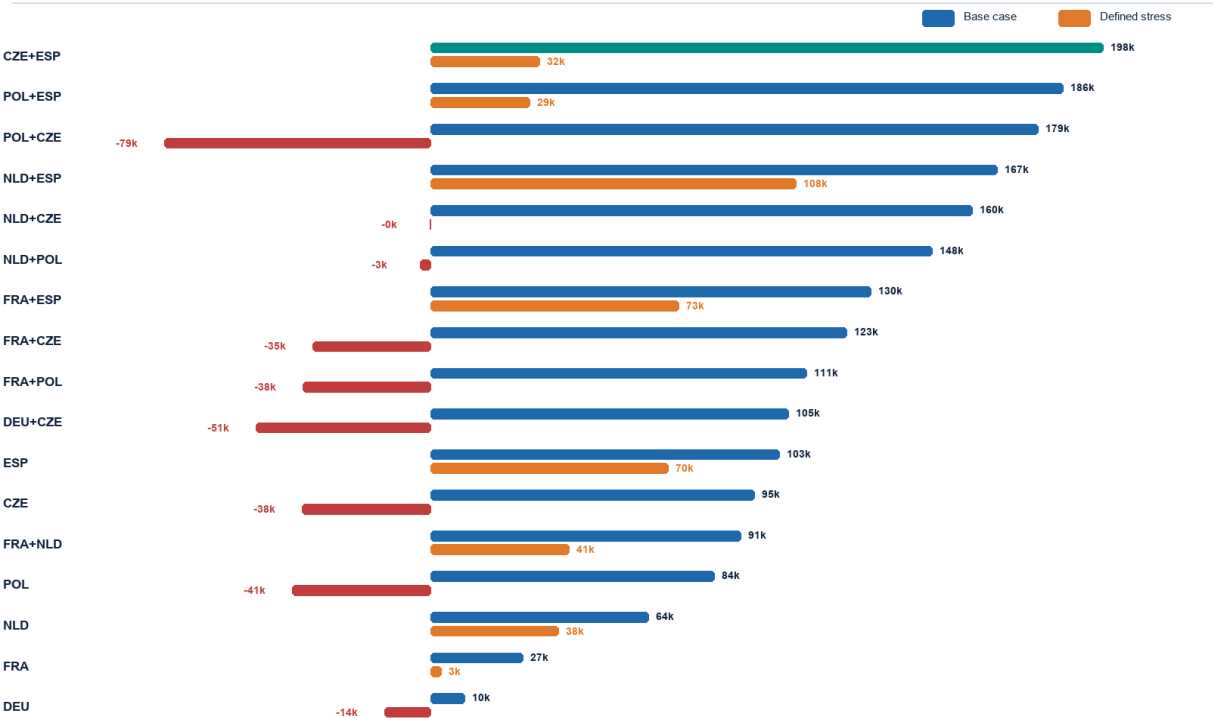
The workbook and metrics.json preserve all twelve monthly PLN and CZK means. The FX series is an official reference-rate archive, not a record of Meridian's actual transaction conversion or a forecast.

6. Options and trade-offs

Scenario policy [S02] defines annual incremental contribution as $C \times u + U \times (1+u) \times s - F$ for low/base/high uplifts $u=\{10\%, 25\%, 40\%\}$; capex K is year-zero and payback is K divided by positive annual incremental contribution. Stress uses the base uplift, a 3% of gross-sales refund shock and a 10% of net-sales FX shock in PLN/CZK markets, with no additional cost recovery. Pairs add country figures without synergy.

Feasible hub options: base contribution with defined stress

Annual incremental contribution, sorted by base case, EUR



Stress = 3% of gross sales refund shock plus 10% of net sales FX shock for PLN/CZK markets; source S02.

Feasible-pair comparison

Pair	Base annual	Capex	FTE	Payback	Stress annual
CZE+ESP	€198,099	€225,000	5	1.14	€32,180
POL+ESP	€186,329	€240,000	6	1.29	€29,264
POL+CZE	€178,995	€205,000	5	1.15	€-78,570
NLD+ESP	€166,931	€270,000	6	1.62	€107,630
NLD+CZE	€159,598	€235,000	5	1.47	€-203
NLD+POL	€147,828	€250,000	6	1.69	€-3,119
FRA+ESP	€129,849	€340,000	7	2.62	€73,105
FRA+CZE	€122,516	€305,000	6	2.49	€-34,729
FRA+POL	€110,745	€320,000	7	2.89	€-37,645
DEU+CZE	€105,472	€375,000	7	3.56	€-51,410
FRA+NLD	€91,348	€350,000	7	3.83	€40,721

The selected CZE+ESP pair is the base-case leader. It gives up some stress resilience versus NLD+ESP, but it has materially higher base contribution and a lower capex requirement. POL+ESP is the strongest base alternative and is close in stress performance; it is the most relevant like-for-like challenge to the recommendation.

Switching conditions: reopen the funding decision if the measured uplift is below 2.74% on the selected pair, realized savings per unit do not cover recurring fixed cost, service-level improvement cannot be verified, or the defined stress becomes negative. These conditions are decision gates, not claims about probabilities.

7. Staged 90-day implementation

Owners below are operating roles, not named people. The sequence protects the board from committing year-zero capex before the synthetic assumptions are checked against live operational evidence.

Timing	Owner	Work / dependency	Decision gate / output
Days 0–15	COO + CFO	Authorize discovery envelope; confirm CZE/ESP lane map, current central-hub service baseline, candidate partner model and capex approval path. Dependency: Finance baseline and Ops data extracts.	Gate 0: approve validation scope only; no full capex release. Output: signed KPI definitions and data owners.
Days 16–45	Ops + Procurement	Validate lane-level freight, handling, labor/partner quotes, local service promise, return routing and staffing plan. Dependency: carrier/partner bids and HR/legal review.	Gate 1: proceed only if validated annual savings and capacity support the scenario, and no hard capex/FTE breach. Output: business case v2.
Days 46–75	Country launch leads	Run controlled operating pilot in the two selected markets with weekly dashboard; reconcile shipped volume, delivery promise, return rate, savings per unit and fulfillment cost against central-hub baseline.	Gate 2: continue only if observed uplift is at least 2.74% run-rate or board explicitly accepts a lower hurdle with evidence. Output: pilot readout.
Days 76–90	Board + COO	Review pilot, finalize contracts, staffing and inventory/returns controls; release capex in tranches if evidence supports the decision.	Gate 3: launch, defer, or switch to POL+ESP/NLD+ESP. Output: signed go/no-go and 13-week operating plan.

Proposed KPI plan

KPI	Frequency / owner	Target or gate
Incremental shipped volume uplift	Weekly / Finance + Ops	≥2.74% run-rate for selected pair before full release; compare with matched central-hub baseline.
Savings per shipped unit	Weekly / Procurement	At or above the option assumption: CZE €2.10; ESP €3.00; report realized and contracted separately.
Contribution per order	Weekly / Finance	Non-negative after refunds, net COGS and fulfillment; reconcile to monthly ledger definition.
Return/refund rate	Weekly / Quality	No deterioration versus 2025 baseline without a documented service or mix explanation.
Promised-day hit rate and cycle time	Weekly / Country leads	Baseline in days 0–15; board sets service threshold after lane validation.
Capex and FTE burn	Weekly / CFO + HR	≤€225k and ≤5 FTE for selected pair; stage release and hiring commitments.

8. Risks, limitations and evidence gaps

Demand risk: 2025 shipped sales are observed history, not a causal forecast of volume uplift from local hubs. Mitigation is a staged pilot and an explicit volume break-even gate.

Economics risk: savings per unit, fixed cost and capex are synthetic policy inputs; local quotes and freight-lane costs are not in the source room. Mitigation is Procurement validation before Gate 1 and reporting contracted versus realized savings.

FX/returns risk: the ECB rate is a reference-rate mean and stress is deliberately conservative but not a probability-weighted forecast. Returns after the cutoff and the orphan return are excluded from the base. Mitigation is weekly return/refund and FX monitoring.

Execution risk: the analysis does not include local regulatory, tax, labor, lease, inventory-transfer or customer service design evidence. These are dependencies for the 90-day plan, not hidden assumptions in the baseline.

Decision strength: the recommendation is supported by reconciled transaction arithmetic and transparent scenario trade-offs. It is not statistically proven, causal or certain. The smallest useful next evidence is the lane-level cost and service validation that can confirm or reject the selected pair's savings and uplift assumptions.

Source register and raw evidence

ID	File	Status	Units	Period	Original URL (truncated)
S01	data-dictionary.md	synthetic_client	see source file	2025 analysis rules	http://127.0.0.1:49807/data-dictionary.md
S02	scenario-policy.md	synthetic_client	see source file	2025 baseline / annual scenario	http://127.0.0.1:49807/scenario-policy.md
S03	orders-part1.csv	synthetic_client	see source file	2025-01 through 2026-01 rows	http://127.0.0.1:49807/orders-part1.csv
S04	orders-part2.csv	synthetic_client	see source file	2025-01 through 2026-01 rows	http://127.0.0.1:49807/orders-part2.csv
S05	order-corrections.csv	synthetic_client	see source file	2025 rows; revisions	http://127.0.0.1:49807/order-corrections.csv
S06	returns.csv	synthetic_client	see source file	2025-01 through 2026-02-01 receipts	http://127.0.0.1:49807/returns.csv
S07	unit-costs.csv	synthetic_client	see source file	2025-01-01 onward	http://127.0.0.1:49807/unit-costs.csv
S08	hub-options.csv	synthetic_client	see source file	Annual recurring / year-zero capex	http://127.0.0.1:49807/hub-options.csv
S09	population.json	public_official_snapshot	persons / current US\$ per person	2022-2024; persons	https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator
S10	gdp-per-capita.json	public_official_snapshot	persons / current US\$ per person	2022-2024; current US\$ per person	https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator
S11	ecb-history.csv	public_official_snapshot	local currency units per EUR	Business-day observations through 2026-09-25; PLN/CZK used for 2025 means	https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip
S12	ecb-history.zip	public_official_snapshot	see source file	Same snapshot as extracted CSV	https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip
S13	source-register.json	synthetic_client_metadata	see source file	Retrieval vintage and original URLs	http://127.0.0.1:49807/source-register.json

Full URLs, retrieval times, hashes and vintages are in deliverables/source_register.json and the workbook Source Register sheet. Reproduction instructions are in REPRODUCE.md.